

Digital Design using FPGA

AXI4-Lite Interface

Design, Simulation, Synthesis & Implementation

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Contents

1	Introduction to AXI4-Lite Interface	2
1.1	AXI4-Lite Interface Signals	2
1.2	Handshake Process	3
1.3	Signals Table	3
2	QuestaSim	6
2.1	Simulation Output	6
2.2	Transcript:	9
3	Vivado	10
3.1	RTL Schematic:	10
3.2	Synthesis Results:	11
3.2.1	Schematic:	11
3.2.2	Timing:	12
3.2.3	Utilization:	12
3.3	Implementation:	13
3.3.1	Timing:	13
3.3.2	Utilization:	13
3.3.3	Device:	14

1. Introduction to AXI4-Lite Interface

1.1 AXI4-Lite Interface Signals

The AXI4-Lite interface consists of five channels: Read Address, Read Data, Write Address, Write Data, and Write Response. An AXI4 read transaction using the Read Address and Data channels is shown in figure 1. Similarly an AXI4 write transaction using the Write Address, Data, and Response channels is shown in figure 2. Note that these figures depict burst transfers, which AXI4-Lite is incapable of.

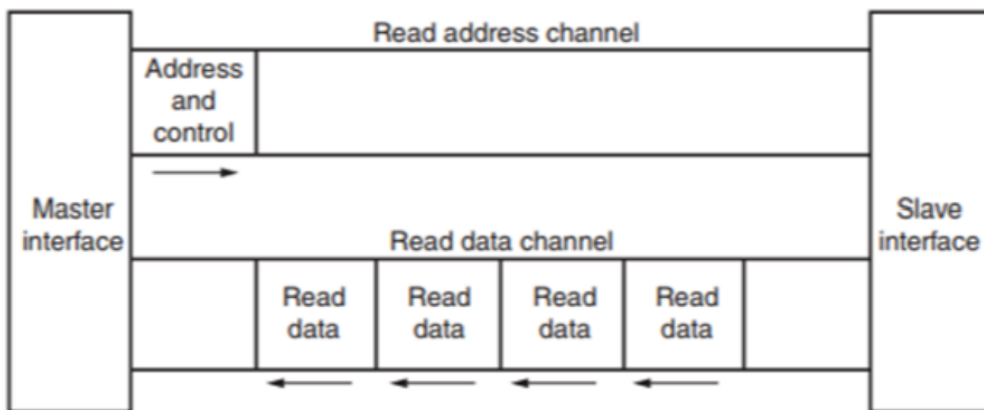


Figure 1.1: Figure 1

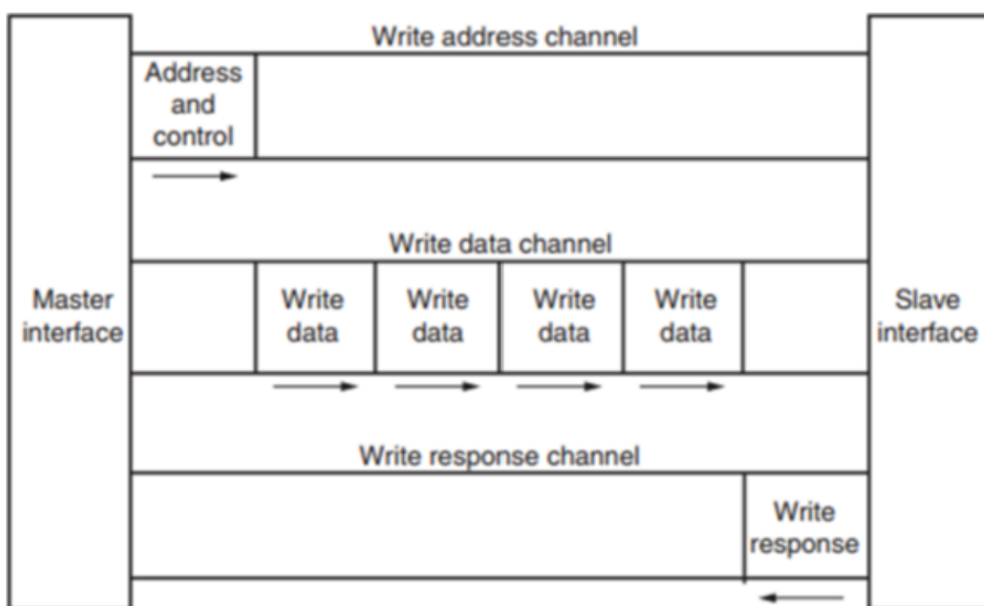


Figure 1.2: Figure 2

1.2 Handshake Process

All five transaction channels use the same VALID/READY handshake process to transfer address, data, and control information. This two-way flow control mechanism means both the master and slave can control the rate at which the information moves between them. The information source generates the VALID signal to indicate when the address, data or control information is available. The information destination generates the READY signal to indicate that it can accept the information. The handshake completes if both VALID and READY signals in a channel are asserted during a rising clock edge.

1.3 Signals Table

Name	Direction	Description
ACLK	$M \rightarrow SM \rightarrow S$	Clock source.
ARESETN	$M \rightarrow SM \rightarrow S$	Global reset source, active low.

Table 1.1: Global Signals

Name	Direction	Description
AWADDR	$M \rightarrow S$	Write address, usually 32-bits wide.
AWVALID	$M \rightarrow S$	Write address valid. Master generates this signal when Write Address and control signals are valid.
AWREADY	$M \rightarrow S$	Write address ready. Slave generates this signal when it can accept Write Address and control signals

Table 1.2: Write Address Channel

Name	Direction	Description
WDATA	$M \rightarrow SM \rightarrow S$	Write data (32-bit only).
WSTRB	$M \rightarrow SM \rightarrow S$	Write strobes. 4-bit signal indicating which of the 4-bytes of Write Data. Slaves can choose assume all bytes are valid.
WVALID	$M \rightarrow S$	Write data valid. Master generates this signal when Write Address and control signals are valid.
WREADY	$M \rightarrow S$	Write data ready. Slave generates this signal when it can accept Write Address and control signals

Table 1.3: Write Data Channel

Name	Direction	Description
ARADDR	$M \rightarrow S$	Read address, usually 32-bit wide.
ARVALID	$M \rightarrow S$	Read address valid. Master generates this signal when Read Address and the control signals are valid.
ARREADY	$M \leftarrow S$	Read address ready. Slave generates this signal when it can accept the read address and control signals.

Table 1.4: Read Address Channel

Name	Direction	Description
RDATA	$M \leftarrow SM \leftarrow S$	Read Data (32-bit only).
RRESP	$M \leftarrow SM \leftarrow S$	Read response. This signal indicates the status of data transfer.
RVALID	$M \leftarrow SM \leftarrow S$	Read valid. Slave generates this signal when Read Data is valid.
RREADY	$M \rightarrow SM \rightarrow S$	Read ready. Master generates this signal when it can accept the Read Data and response.

Table 1.5: Read Data Channel

Name	Direction	Description
BRESP	$M \leftarrow SM \leftarrow S$	Write response. This signal indicates the status of the write transaction.
BVALID	$M \leftarrow SM \leftarrow S$	Write response valid. Slave generates this signal when the write response on the bus is valid.
BREADY	$M \rightarrow SM \rightarrow S$	Response ready. Master generates this signal when it can accept a write response.

Table 1.6: Write Response Channel

2. QuestaSim

2.1 Simulation Output

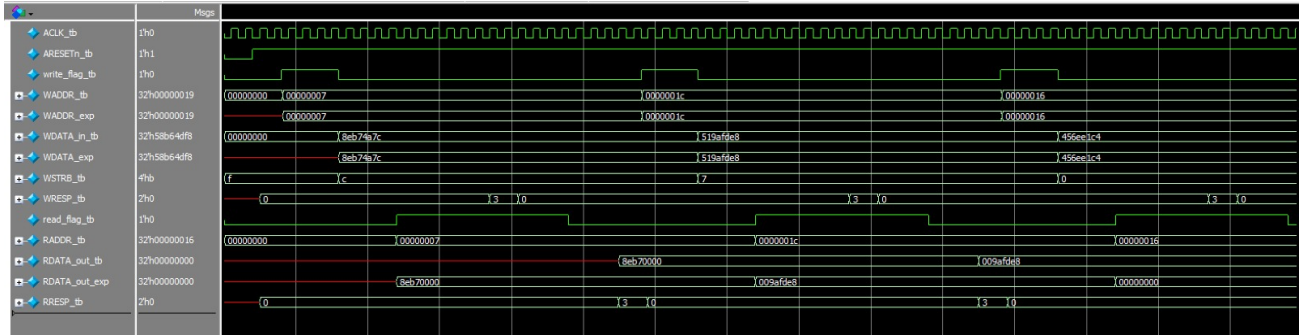


Figure 2.1: Simulation waveform of Top Module testbench (1)

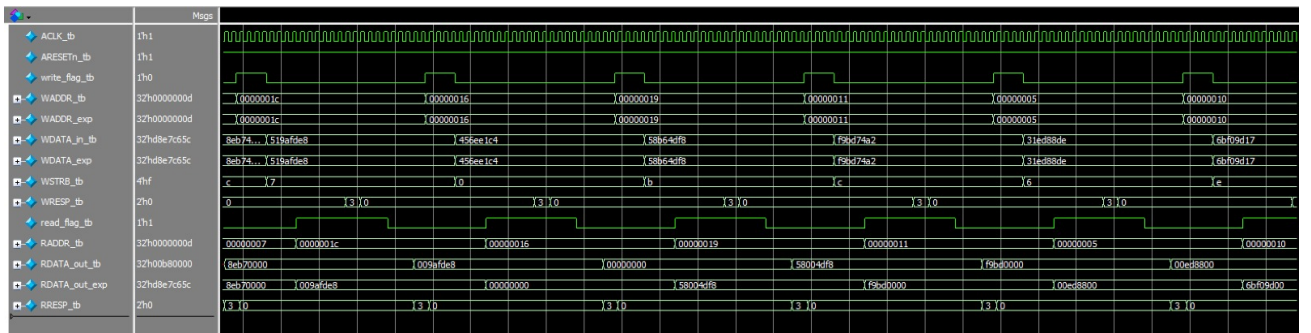


Figure 2.2: Simulation waveform of Top Module testbench (2)

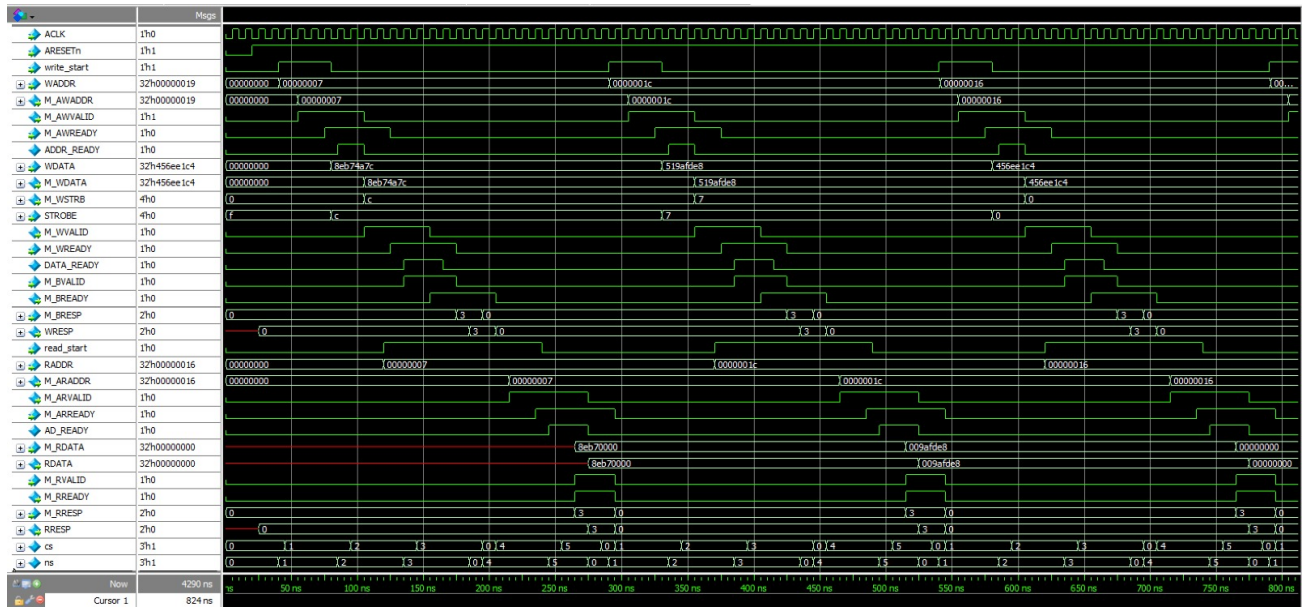


Figure 2.3: Simulation waveform of AXI-Lite Master module (1)

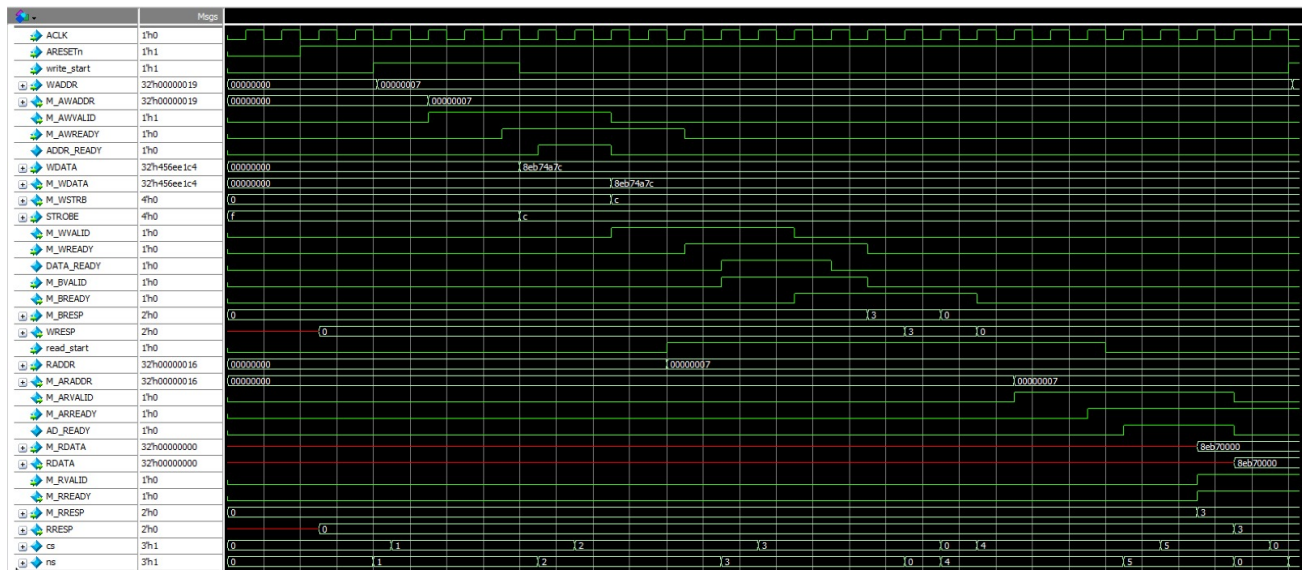


Figure 2.4: Simulation waveform of AXI-Lite Master module (2)

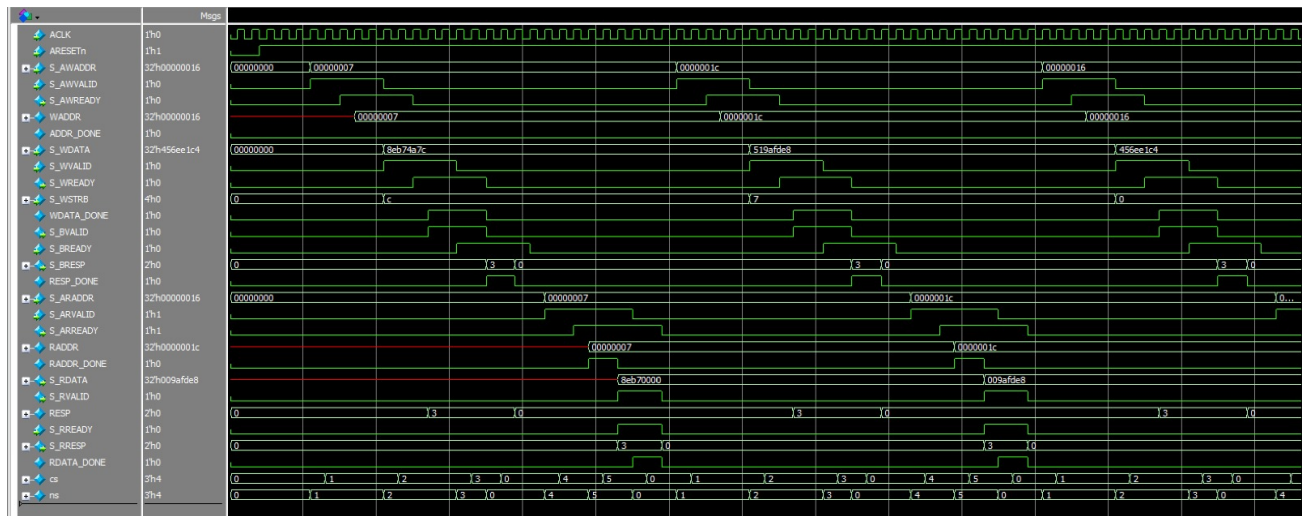


Figure 2.5: Simulation waveform of AXI-Lite Slave module (1)

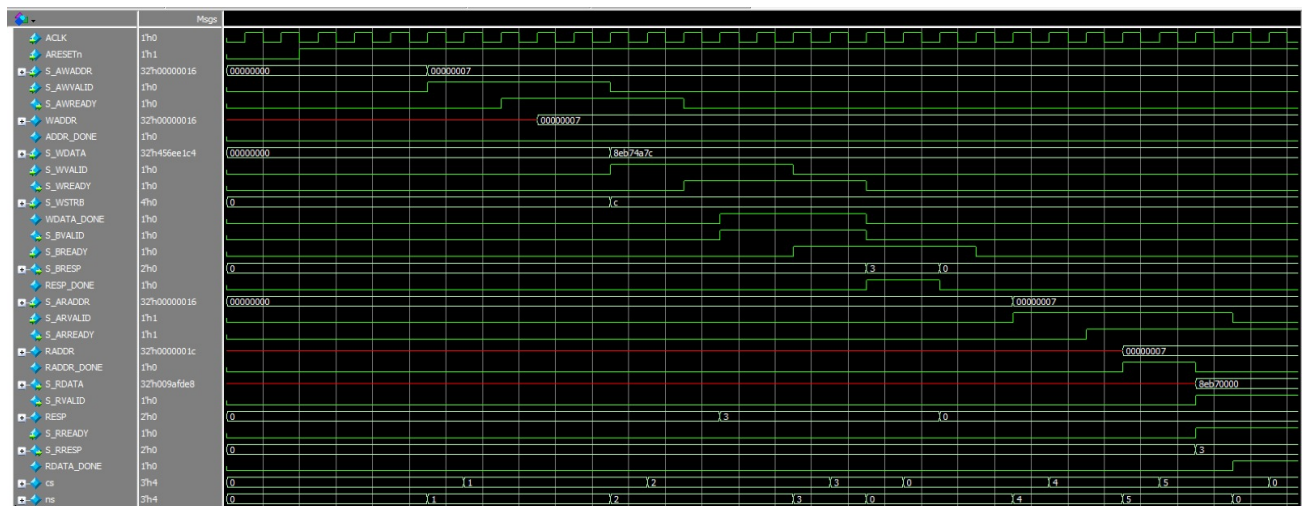


Figure 2.6: Simulation waveform of AXI-Lite Slave module (2)

2.2 Transcript:

```

# CORRECT, ReadDataOUT = 8eb70000, ReadExp = 8eb70000, Address = 00000007, STROBE = c
# CORRECT, ReadDataOUT = 009afde8, ReadExp = 009afde8, Address = 0000001c, STROBE = 7
# CORRECT, ReadDataOUT = 00000000, ReadExp = 00000000, Address = 00000016, STROBE = 0
# CORRECT, ReadDataOUT = 58004df8, ReadExp = 58004df8, Address = 00000019, STROBE = b
# CORRECT, ReadDataOUT = f9bd0000, ReadExp = f9bd0000, Address = 00000011, STROBE = c
# CORRECT, ReadDataOUT = 00ed8800, ReadExp = 00ed8800, Address = 00000005, STROBE = 6
# CORRECT, ReadDataOUT = 6bf09d00, ReadExp = 6bf09d00, Address = 00000010, STROBE = e
# CORRECT, ReadDataOUT = 00b80000, ReadExp = 00b80000, Address = 00000003, STROBE = 4
# CORRECT, ReadDataOUT = d8e7c65c, ReadExp = d8e7c65c, Address = 0000000d, STROBE = f
# CORRECT, ReadDataOUT = c182e897, ReadExp = c182e897, Address = 00000012, STROBE = f
# CORRECT, ReadDataOUT = 000000aa, ReadExp = 000000aa, Address = 00000002, STROBE = 1
# CORRECT, ReadDataOUT = 00000faa, ReadExp = 00000faa, Address = 00000002, STROBE = 2
# CORRECT, ReadDataOUT = 0000de00, ReadExp = 0000de00, Address = 00000006, STROBE = 2
# CORRECT, ReadDataOUT = 0000ff00, ReadExp = 0000ff00, Address = 00000008, STROBE = 2
# CORRECT, ReadDataOUT = 00001234, ReadExp = 00001234, Address = 00000009, STROBE = f
# CORRECT, ReadDataOUT = 00005678, ReadExp = 00005678, Address = 00000005, STROBE = f
# CORRECT, ReadDataOUT = 0000defa, ReadExp = 0000defa, Address = 00000004, STROBE = f
# ** Note: $stop      : C:/Users/LEGION/Desktop/NTIFPGA/NTI_PROJECT/AXI4_lite_tb.v(350)
#   Time: 4290 ns   Iteration: 1   Instance: /AXI4_lite_tb

```

Figure 2.7: Transcript

3. Vivado

3.1 RTL Schematic:

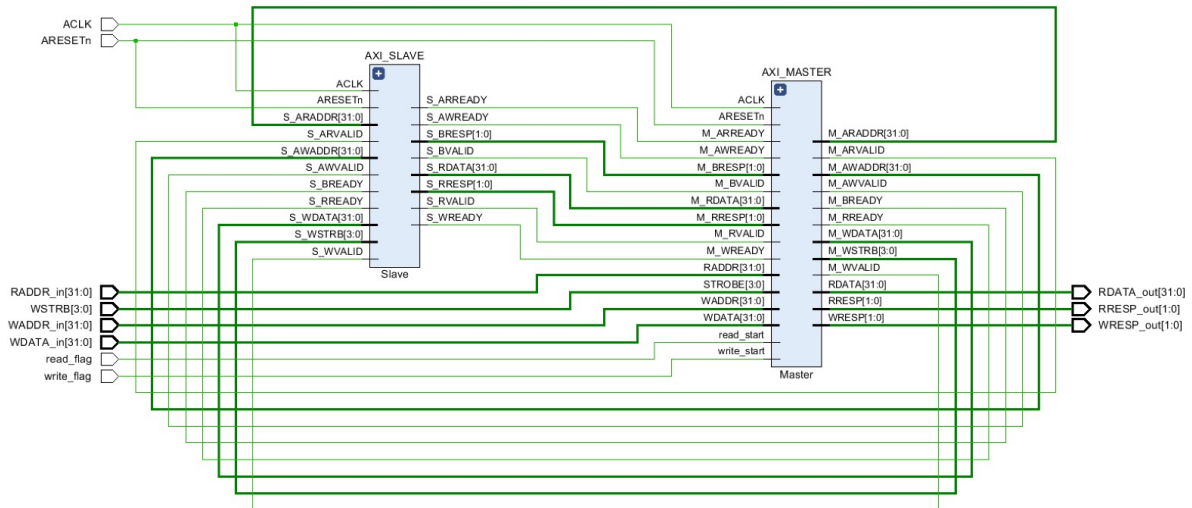


Figure 3.1: Elaborated Design

3.2 Synthesis Results:

3.2.1 Schematic:

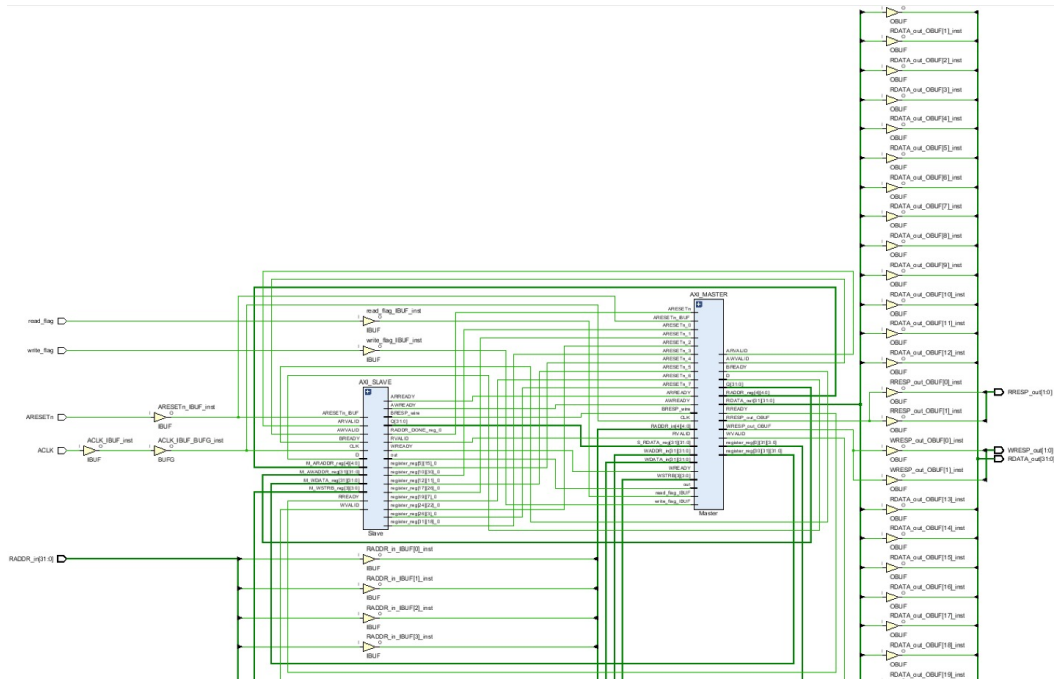


Figure 3.2: Schematic After Synthesis(1)

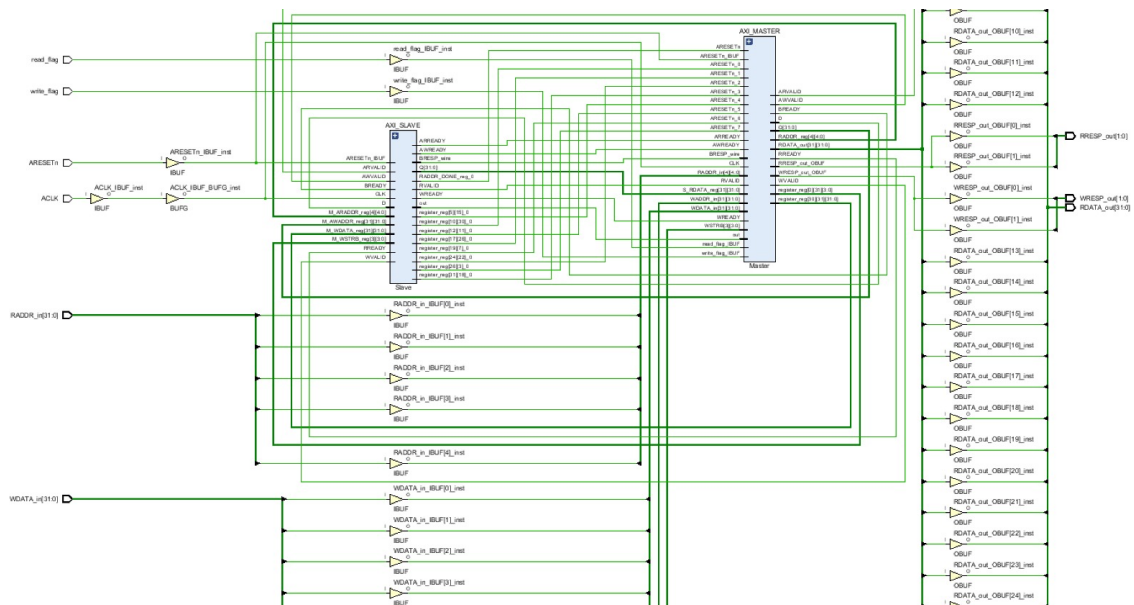


Figure 3.3: Schematic After Synthesis(2)

3.2.2 Timing:

Design Runs	Utilization	Timing	×	Debug
Design Timing Summary				
Setup		Hold		Pulse Width
Worst Negative Slack (WNS): 6.944 ns		Worst Hold Slack (WHS): 0.129 ns		Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns		Total Hold Slack (THS): 0.000 ns		Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0		Number of Failing Endpoints: 0		Number of Failing Endpoints: 0
Total Number of Endpoints: 2371		Total Number of Endpoints: 2371		Total Number of Endpoints: 1233
All user specified timing constraints are met.				

Figure 3.4: Timing Report After Synthesis

3.2.3 Utilization:

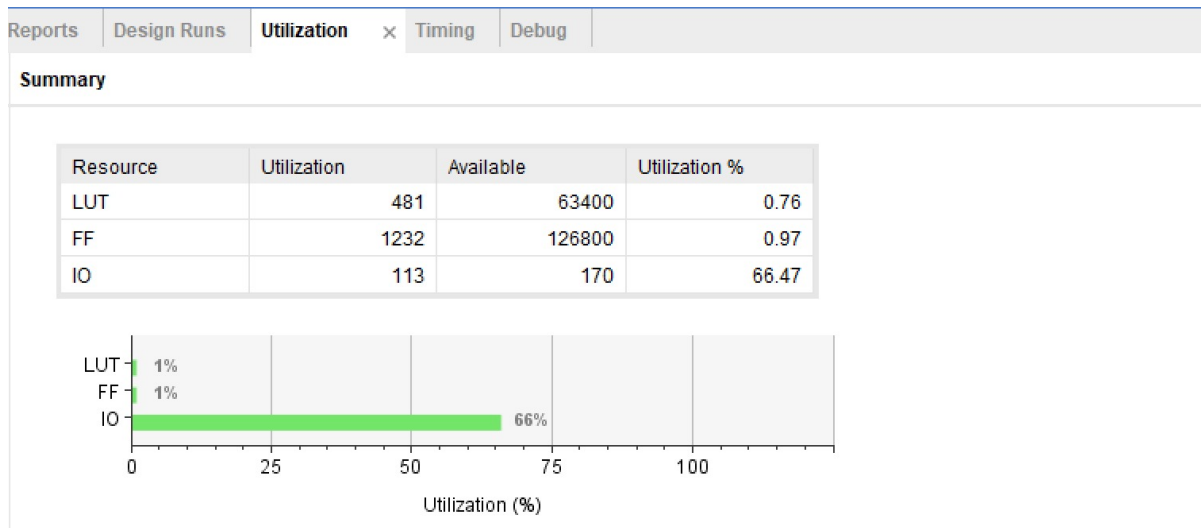


Figure 3.5: Utilization Report After Synthesis

3.3 Implementation:

3.3.1 Timing:

Design Timing Summary			
Setup	Hold	Pulse Width	
Worst Negative Slack (WNS): 3.781 ns	Worst Hold Slack (WHS): 0.084 ns	Worst Pulse Width Slack (WPWS): 4.500 ns	
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns	
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	
Total Number of Endpoints: 2371	Total Number of Endpoints: 2371	Total Number of Endpoints: 1233	
All user specified timing constraints are met.			

Figure 3.6: Timing Report After Synthesis

3.3.2 Utilization:

reports	Design Runs	Power	DRC	Methodology	Timing	Utilization				
				Hierarchy						
Name	1	Slice LUTs (63400)	Slice Registers (126800)	F7 Muxes (31700)	Slice (1585 0)	LUT as Logic (63400)	LUT Flip Flop Pairs (63400)	Bonded IOB (170)	BUFGCTRL (32)	
Top		481	1232	128	509	481	59	113	1	
AXI_MASTER (Master)		19	121	0	44	19	13	0	0	
AXI_SLAVE (Slave)		462	1111	128	473	462	45	0	0	

Figure 3.7: Utilization Report After Implementation

3.3.3 Device:

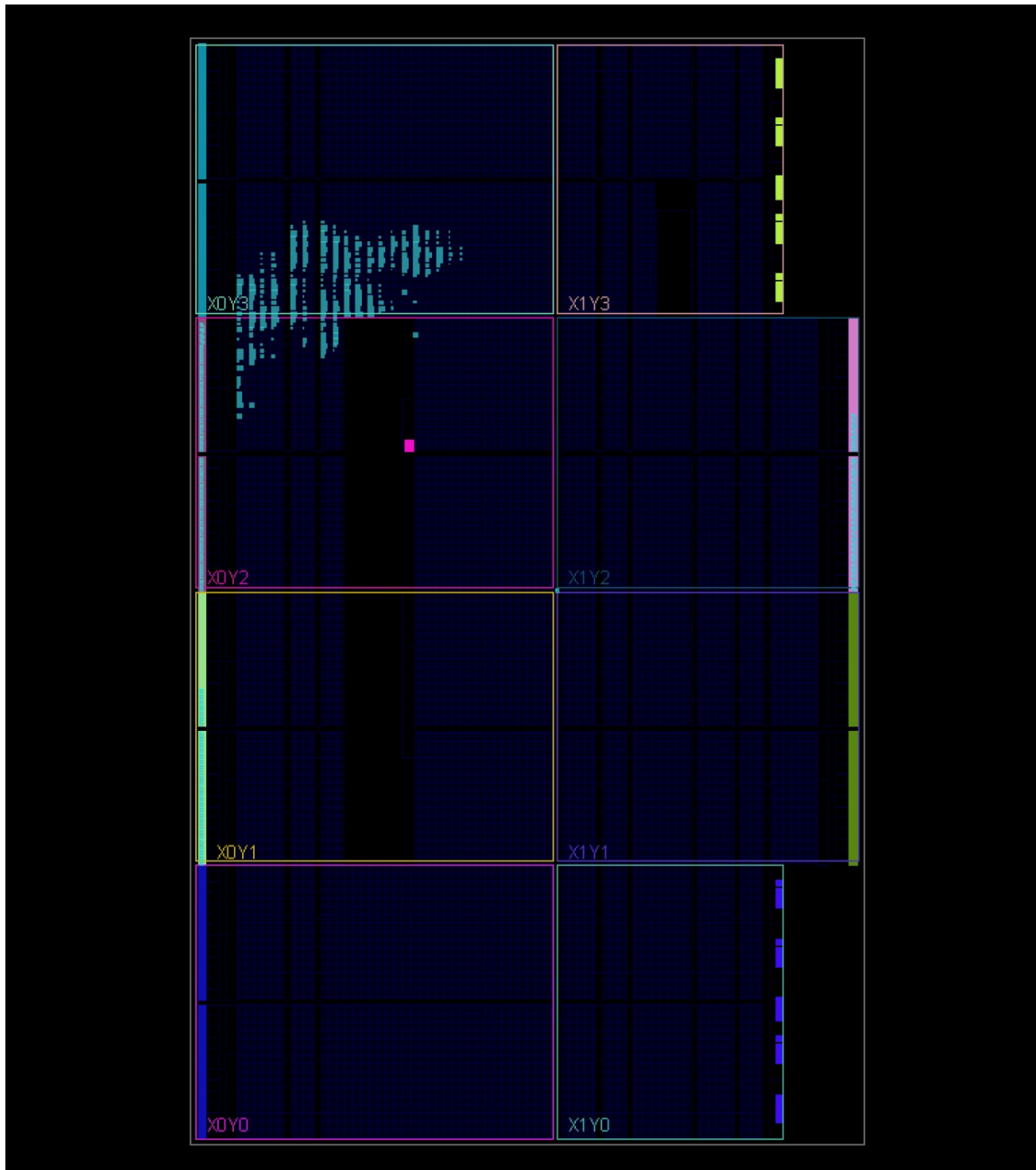


Figure 3.8: Implemented Device